Exercise 9: Gradient Descent and Hyperparameter Optimization

Task 1:

a) What is the fundamental difference between the model's parameters and hyperparameters?
b) What is the validation set used for? What would happen if samples from the train/test set were used for the same purpose?

Task 2:

The MLP you trained yields very low error on the training data, but after the training is finished, the model's predictions still result in high error on the test data.

- What is this phenomenon called?
- What could cause this problem?
- What methods could you apply to address it?

Task 3:

Revisit the MLP for the MNIST digits dataset from the last exercise sheet. This time, apply a *random search* for finding optimal values for the hyperparameters:

- Number of layers
- Number of nodes per layer
- Learning rate

For each of these hyperparameters, choose 5 reasonable values to search from. In each iteration of random search, randomly select one value for each hyperparameter and use them to train and evaluate the MLP. Repeat this process for 10 iterations.

Split 20% of data from the training set as the validation set and test which iteration performs best on the validation data. Report on this iteration's performance on the test set and compare it to your model from last week.

Optionally: Use this validation data to apply *early stoppage* if no improvements on the validation loss has been made for 5 epochs.

Task 4:

Inspect the following error/cost-parameter curve. Mark regions that illustrate typical optimization challenges encountered during backpropagation training. For each marked region, name the issue, explain why it occurs, and discuss how it affects convergence.